

18. (1 point)

a. (0.5 point)

Briefly describe two difficulties when using the Equity as a Call Option method to value a P&C insurance company.

b. (0.5 point)

Discuss a common difficulty when using the Dividend Discount Model, Discounted Cash Flow, and Abnormal Earnings methods to value a P&C insurance company.

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SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 18	
TOTAL POINT VALUE: 1	LEARNING OBJECTIVE(S): B2
SAMPLE ANSWERS	
Part a: 0.5 point	
<u>Sample Responses</u> Any combination of two of the sample answers below. • It's hard to distinguish between debt and policyholder liabilities. • Policyholder liabilities do not have a single expiration date. • Often priced using Black-Scholes model. Black-Scholes inputs like risk free rate, volatility, time horizon is hard to estimate.	
Part b: 0.5 point	
<u>Sample Responses</u> Varying combination of these concepts received credit. • These methods require forecasts of financial variables which is hard as we do not know the company's plan for growth nor do we have detailed information for forecasting. • It may be difficult to calculate the terminal value since it's difficult to project the company's future earnings. • All of these methods require discounting cash flows to a single point in time. Must decide between using firm beta to calculate k which may be volatile or use industry beta which may not perfectly reflect the risk profile of any given P&C company. K equals discount rate based on CAPM.	
EXAMINER'S REPORT	
Candidates were expected to understand the Dividend Discount Model, Discounted Cash Flow, Abnormal Earnings and Equity as a Call Option methods to value a P&C insurance company and be able to discuss the difficulties of using each method.	
Part a	
Candidates were expected to provide two valid difficulties when using the Equity as a Call Option. Common mistakes included: • Providing answers which were not specific enough. • Discussing difficulties with Real Options Valuation rather than the Equity as a Call Option method.	
Part b	
Candidates were expected to discuss a valid difficulty that is common to the three methods listed in the question. Common mistakes included: • Providing answers which were not specific enough, such as saying the methods are complex, difficult to parameterize or sensitive to inputs without further explanation. • Providing difficulties that were not common to all three methods.	